

Centering Update

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Abstract

I propose a theory of updating on centered propositions. According to the theory, agents should adopt credences which—among all credences that conform to the relevant evidence—are closest to their original credences. As I explain, this theory draws connections between accuracy, imaging, and indifference. It generalizes to cases where the relevant evidence includes information about chances. And it enjoys several advantages over alternative updating theories.

1 Introduction

After waking up dazed, Ansel cannot tell whether it is Wednesday, Thursday, Friday, or Saturday. Perhaps Ansel has credence $\frac{1}{4}$ in each of these possibilities. Then Ansel sees a text from Gita, saying that she cannot wait for the weekend. Because of how Gita phrased her text, Ansel learns the centered proposition that it is not Saturday. So Ansel’s new evidence rules out the Saturday possibility. And that raises a question: how should Ansel’s credences in the other possibilities change? More generally, how should agents update their credences after getting ‘centered evidence’, that is, evidence containing centered propositions?

To answer these questions, I propose a principle of rationality—call it ‘Close Update’—for updating on centered propositions like “It is not Saturday.” Close Update embodies an accuracy-first approach to rationality (Briggs & Pettigrew,

2020; Joyce, 2009; Pettigrew, 2016). For it invokes an attractive account of what it is for one credence function to be ‘as close as possible’ to another.

As I discuss later, Close Update enjoys some advantages over alternative theories of updating on centered propositions (Meacham, 2008; Moss, 2012; Titelbaum, 2008). Those other theories generally privilege uncentered propositions in various ways, treating centered propositions as secondary or special cases. That is indicative of a larger trend, in the literature, to approach centered propositions as if they were relatively minor anomalies. But that trend is problematic. Centered propositions are neither anomalous nor secondary to uncentered propositions. We learn centered propositions all the time: arguably, more often than we learn uncentered propositions. Check the time, and you learn the centered proposition “It is thus-and-so o’clock.” Self-reflect, and you learn the centered proposition “I am thus-and-so.” Answer the door, and you learn the centered proposition “Thus-and-so is here.” So there is need for a theory of updating which treats centered propositions as primary rather than secondary. And that is what Close Update delivers: in effect, this paper advocates a gestalt shift in how to approach questions of rationality, by suggesting that centered propositions take center stage.

Importantly, Close Update fills a gap in the literature concerning exactly how accuracy, imaging, and indifference relate. For Close Update is equivalent to an update procedure based on imaging, which in turn is equivalent to an update procedure based on indifference of a qualified form. As I mention later, the proofs of these equivalences complement—and in a sense, weaken the assumptions underwriting—work due to Leitgeb and Pettigrew (2010a, 2010b). So Close Update forges striking connections between (i) centered propositions, (ii) accuracy, (iii) imaging, and (iv) indifference. In addition, on top of that, Close Update supports a natural extension of those connections to situations involving chance: an extension which, in turn, can be used to defend the popular third view in Elga’s case (2000), and to derive the intuitive verdicts in the Judy Benjamin case (van Fraassen, 1981).

So altogether, Close Update provides a unified, rigorous, and comprehensive account of updating on centered evidence. Accuracy, imaging, indifference, and chance, find unity in cases involving centered propositions. This cohesion and

theoretical harmony, which Close Update exhibits, is noteworthy.

In Section 2, I formulate Close Update, provide arguments in its favor—including arguments based on imaging, indifference, and chance—and defend it against objections. In Section 3, I argue that Close Update is better than alternative updating principles in the literature, and I explain the precise sense in which Close Update challenges the problematic tendency to treat centered propositions as anomalous. An appendix contains proofs of new theorems which those arguments invoke.

2 Close Update

This section focuses on Close Update. In Section 2.1, I present Close Update itself, along with examples of its intuitive verdicts. In Section 2.2, I present two other principles—based on imaging and indifference—with which Close Update is equivalent, and which I use to argue for Close Update; then I present another principle which supports arguments for Close Update as well. In Section 2.3, I defend Close Update against objections. Finally, in Section 2.4, I extend Close Update to a principle which covers situations involving chance, and I use that principle to derive standard verdicts in Elga’s case and the Judy Benjamin case.

2.1 The Principle

Roughly put, Close Update is as follows. Suppose that an agent, with an initial credence function, learns some centered propositions. Then the agent should shift to a new credence function which has two properties. First, the new credence function conforms to the centered propositions that the agent learned. Second, among all credence functions which conform to the centered propositions that the agent learned, the new credence function is closest to the agent’s initial credence function.

Some formalism will be helpful. Let $\mathcal{W}$ be a finite set of centered worlds.¹

¹I often refer to elements of $\mathcal{W}$ as ‘worlds’ rather than ‘centered worlds’.

As usual, model centered propositions as sets of centered worlds, and model centered worlds as tuples consisting of (i) various ‘centers’, like times and agents and locations, and (ii) an uncentered world (Kaplan, 1989; Lewis, 1979; Ninan, 2013).² For brevity, I use one and the same symbol ‘ w ’ to represent both a world and the unique proposition corresponding to that world. Let $\mathcal{C}$ be the set of probabilistic credence functions over the power set of $\mathcal{W}$. Since each credence function in $\mathcal{C}$ is determined by its values on $\mathcal{W}$ alone, in what follows, I abuse notation and take $\mathcal{C}$ to contain functions over $\mathcal{W}$ rather than over the powerset.

The main tool from the accuracy literature, which Close Update employs, is a function that quantifies degrees of closeness between credences. This function, called the ‘squared Euclidean semimetric’, generates a topology that represents a natural notion of closeness on $\mathcal{C}$ (Willard, 1970, pp. 173-174); and that topological notion of closeness is all I need, to formulate Close Update. There are many reasons to focus on the squared Euclidean semimetric in this paper: it is used throughout the accuracy literature to define the Brier score, for instance; it is also more-or-less the unique way of quantifying closeness between credences which has a series of attractive formal properties.³ For arguments in favor of using the squared Euclidean semimetric, see (Briggs & Pettigrew, 2020; Easwaran, 2013; Joyce, 2009; Leitgeb & Pettigrew, 2010b; Pettigrew, 2016).⁴

Other functions, which quantify degrees of closeness between credences, could be used instead of the squared Euclidean semimetric. Some formal results established in this paper do not hold for those other functions, while other formal results do: for instance, some such results do not hold for the Kullback-Leibler divergence. The reader is welcome to adopt their own preferred function for quantifying degrees of closeness between credences. They may just need to supply their own versions of a few formal arguments to come.

²The centered proposition that I am Ansel, for instance, is the set of all centered worlds $\langle c, w \rangle$, where c is Ansel and w is an uncentered world.

³Up to a constant multiple, it is symmetric, and it is definable in a canonical way—based on first-order Taylor expansions—using a continuously differentiable, strictly convex function. These formal properties generalize the properties, like being strictly proper, upon which philosophers focus.

⁴Many of these arguments ultimately justify the Brier score; and as part of doing that, they justify using the squared Euclidean semimetric to quantify degrees of closeness between credences (Pettigrew, 2016, p. 8).

Three more definitions are needed. Here is the first: an agent's 'total centered world evidence' is the set of all worlds in $\mathcal{W}$ which, for that agent, are epistemically possible. In general, I use ' E ' to represent a given agent's total centered world evidence.

An important aside: as I discuss later, there are cases where an agent's total centered world evidence is merely one part of the agent's total evidence. In many standard cases, the agent's total centered world evidence just is their total evidence – and Close Update describes how agents should update in cases like that. But in other cases, the agent's total evidence does more than simply classify some worlds as epistemically possible and other worlds as epistemically impossible. The agent's total evidence includes information about what the agent remembers, for instance, or information about chances. In cases like that, Close Update does not apply; a modification of it, however, does. I discuss this later.

For the second definition, let c be a credence function in $\mathcal{C}$, and let E be a subset of $\mathcal{W}$. Say that c 'conforms to E ' just in case for all centered worlds w in $\mathcal{W}$ which are not in E , $c(w) = 0$. In other words, an agent's credences conform to some total centered world evidence just in case that agent has zero credence in all worlds which that evidence rules out.

For the third definition, let c and c' be credence functions in $\mathcal{C}$, and let E be a subset of $\mathcal{W}$. Then say that c' is the 'closest credence function to c which conforms to E ' just in case for all credence functions c'' in $\mathcal{C}$ which conform to E , the degree of closeness between c and c'' is at least as great as the degree of closeness between c and c' . Note that all these 'degrees of closeness' are quantified using the squared Euclidean semimetric.⁵

Now for Close Update. Basically, it says that after receiving some centered evidence, an agent should update to the closest credence function which that evidence does not rule out.

Close Update

Let A be an agent. Let t be a time, and let t' be the time immediately

⁵Since the set of credence functions which conform to any given E is closed and convex, there always exists a unique closest credence function to c in that set.

after t .⁶ Let c and c' be the credence functions which A has at t and t' , respectively. Suppose that at t' , A 's total evidence is some total centered world evidence E . Then in order for A to be rational, c' should be the closest credence function to c which conforms to E .

In other words, suppose that your total evidence is just your total centered world evidence. Then in order to rationally update your credences in response to that evidence, do the following. Consider all the credence functions which assign zero to the worlds that your evidence rules out. Among those credence functions, find the one closest to the credence function which you just had. According to Close Update, that is the credence function which you should adopt.

Before discussing examples, it is worth briefly reviewing a key theorem—namely, theorem 1, proved in the appendix—which helps convey the intuitive content of Close Update. As the theorem shows, upon obtaining new total centered world evidence, Close Update is equivalent to the following. First, assign zero credence to the worlds which that evidence rules out. Second, sum up your previous credences in those ruled-out worlds. Third, divide the summed credences among all the worlds compatible with your new total centered world evidence. As shown by the theorem, the credence function generated by this procedure is identical to the credence function which Close Update recommends.

In addition to facilitating various calculations in later examples, this theorem plays another extremely important role in what follows. Specifically, this theorem serves as a linchpin for many different philosophical arguments in favor of Close Update, presented in Section 2.2. One argument relies on the fact that given this theorem, Close Update is equivalent to an attractive update principle based on imaging. Another argument relies on the fact that given this theorem, Close Update is equivalent to an attractive kind of indifference principle. Yet another argument relies on more general considerations concerning the importance of minimal revisions in response to new evidence. So altogether, this theorem plays a crucial role in (i) conveying what Close Update says, (ii) facilitating calculations that support applications of Close Update to particular cases, and (iii) underwriting more general philosophical arguments for Close Update.

⁶For simplicity, assume that time is discrete.

For an example application of Close Update, take the case of Ansel from Section 1. In that case, there are four centered worlds: “It is Wednesday,” “It is Thursday,” “It is Friday,” and “It is Saturday.” Initially, after waking, Ansel assigns credence $\frac{1}{4}$ to each of them. After reading Gita’s text, Ansel’s total evidence becomes the following: it is not Saturday; in other words, it is either Wednesday, Thursday, or Friday. Then as implied by theorem 1 in the appendix, according to Close Update, Ansel’s new credence function should (i) assign 0 to “It is Saturday,” and (ii) assign $\frac{1}{3}$ to each of “It is Wednesday,” “It is Thursday,” and “It is Friday.” And that is the intuitively correct result; so this is a point in Close Update’s favor.

Here is a more complicated application of Close Update. Consider the following three worlds.

1. d_1 : dugongs eat seagrass and it is January 1 of 2025.
2. $(\neg d)_1$: dugongs do not eat seagrass and it is January 1 of 2025.
3. d_2 : dugongs eat seagrass and it is January 2 of 2025.

Suppose that initially, Jie’s credences are such that $c(d_1) = \frac{1}{3}$, $c((\neg d)_1) = \frac{2}{3}$, and $c(d_2) = 0$. In addition, suppose that Jie’s total evidence—which is just Jie’s total centered world evidence, in this case—becomes $\{(\neg d)_1, d_2\}$. Then as implied by theorem 1 in the appendix, according to Close Update, Jie’s new credences should be such that $c'(d_1) = 0$, $c'((\neg d)_1) = \frac{5}{6}$, and $c'(d_2) = \frac{1}{6}$.

A nice feature of this: Close Update allows Jie’s credence in d_2 to (i) be zero, and then (ii) shift to a non-zero value. That is an advantage of Close Update over more standard theories of updating. Conditionalization, for instance, implies that if an agent has credence zero in a proposition at one time, then that agent is rationally required to have credence zero in that proposition at all later times. There is more to say about this issue; I discuss it in more detail in Section 2.3. But for now, just note that as the case involving Jie demonstrates, for many propositions, this implication is problematic. It makes sense that after obtaining total centered world evidence $\{(\neg d)_1, d_2\}$, Jie’s credence in d_2 should increase from zero to something non-zero. Close Update respects this, whereas conditionalization does not.

Close Update can be used to produce the intuitively correct verdicts in less abstract, and correspondingly more complex, cases as well. To see how, it is worth considering the evolution of credences in the following realistic case – note that to

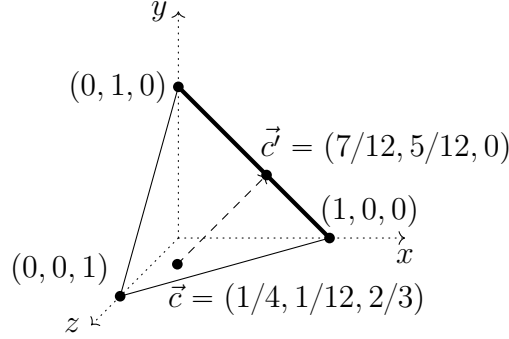
avoid unnecessarily complex quantitative details, I focus on how the credences in this case evolve qualitatively.⁷ Suppose you pour water on some tea leaves, and then wait for the right time at which to remove those leaves from your cup. You do not know the current time exactly, but you have beliefs about the correlation between the color of the tea leaves and the amount of time which has passed: so you believe various propositions of the form “the tea leaves have such-and-such shade of color if and only if it has been thus-and-so long since the water was poured.” At each moment, you also have credences about approximately how long ago you poured the water. Finally, at each moment, you are completely certain that if it has been two minutes since the pouring, then it is time to remove the tea leaves.

Given all this, Close Update reproduces the intuitively correct verdict regarding when, in your estimation, the tea leaves should be removed. Initially, you have high credence in the proposition that two minutes have not yet passed: perhaps you are certain of this because you remember how long it has been; or because you can tell, from the color of the tea leaves, that it has not been two minutes yet. So shortly after pouring the water, as time passes from one moment to the next, the closest credence function to your credence function at the prior moment—which is compatible with your total centered world evidence—also assigns extremely high credence to the proposition that two minutes have not yet passed: for your evidence does not rule this proposition out, and so your credence in this proposition remains relatively constant. Eventually, however, you become pretty certain that it has been around two minutes since the pouring; this proposition becomes part of your total centered world evidence. At that time, the closest credence function to your credence function at the prior moment, which is compatible with your evidence, assigns high credence to the proposition that the water was poured two minutes ago. Because you are also completely certain that if it has been two minutes then it is time to remove the tea leaves, it follows that you have extremely high credence in the centered proposition that it is time for the removal. Therefore, given the temporal evolution both of your memory about the pouring and of the leaves’ color, Close Update implies that your credences evolve in the intuitively correct way.

⁷Thanks to an anonymous reviewer for this case.

Before continuing, it is worth discussing an interesting technical aside: Close Update admits of a lovely geometric interpretation, along the lines discussed in (Dorst et al., 2021). A credence function, according to this interpretation, is a vector in a special space. Total centered world evidence is a special subspace within that space. And Close Update can be re-expressed, equivalently, like this: after agent A —with initial credences represented by vector $\vec{c}$ —obtains total centered world evidence represented by subspace E , A ’s new credences $\vec{c}'$ should be the projection of $\vec{c}$ into the E subspace.

Though the formal details of this interpretation are somewhat involved,⁸ the basic idea can be illustrated quite easily for the case where $\mathcal{W} = \{w_1, w_2, w_3\}$. Towards that end, consider the picture below.



The triangle is called a ‘2-simplex’; this is the ‘special space’ mentioned earlier. Each vector in that 2-simplex represents a credence function: $(1, 0, 0)$ represents the credence function which assigns 1 to w_1 and 0 to the other worlds, $(0, 1, 0)$ represents the credence function which assigns 1 to w_2 and 0 to the other worlds, and $(0, 0, 1)$ represents the credence function which assigns 1 to w_3 and 0 to the

⁸Here are the formal details. For each $c \in \mathcal{C}$, let $(c(w_1), c(w_2), \dots, c(w_n)) =_{df} (c_1, c_2, \dots, c_n) =_{df} \vec{c}$ be an element of the closed, convex set $\Delta^{n-1} =_{df} \{\vec{x} = (x_1, \dots, x_n) \in \mathbb{R}^n \mid x_1, \dots, x_n \in [0, 1] \text{ and } \sum_{i=1}^n x_i = 1\}$. Note that while $\mathbb{R}^n$ is a vector space, Δ^{n-1} is not; instead, Δ^{n-1} is an $(n-1)$ -simplex, basically an $(n-1)$ -dimensional generalization of a triangle or a tetrahedron. The squared Euclidean semimetric $\mathfrak{s}$ is defined like this: for all $\vec{c}, \vec{c}' \in \mathbb{R}^n$, $\mathfrak{s}(\vec{c}, \vec{c}') = \sum_{i=1}^n (c_i - c'_i)^2$. For each total centered world evidence $E = \{w_{\sigma_1}, \dots, w_{\sigma_k}\} \subseteq \mathcal{W}$, define the $(k-1)$ -simplex $\Delta_E^{k-1} = \{\vec{c} = (c_1, c_2, \dots, c_n) \in \Delta^{n-1} \mid \forall i \notin \{\sigma_1, \dots, \sigma_k\}, c_i = 0\}$. Finally, define the semimetric projection $P_E : \Delta^{n-1} \rightarrow \Delta_E^{k-1}$ as follows: for all $\vec{c} \in \Delta^{n-1}$, $P_E(\vec{c}) = \min_{\vec{x} \in \Delta_E^{k-1}} \mathfrak{s}(\vec{c}, \vec{x})$; that is, $P_E(\vec{c})$ is whichever $\vec{x} \in \Delta_E^{k-1}$ minimizes $\mathfrak{s}(\vec{c}, \vec{x})$. Note that since Δ_E^{k-1} is closed and convex, P_E is well-defined. For more on projections like P_E , see (Phelps, 1993, p. 8; Polyak, 2021, p. 36).

other worlds. The vector $\vec{c}$ represents the credence function which assigns $\frac{1}{4}$, $\frac{1}{12}$, and $\frac{2}{3}$ to w_1 , w_2 , and w_3 respectively. And the vector $\vec{c'}$ represents the credence function which assigns $\frac{7}{12}$, $\frac{5}{12}$, and 0 to w_1 , w_2 , and w_3 respectively. The bolded line in the $x-y$ plane is called a ‘1-simplex’, and in the following way, it represents total centered world evidence $E = \{w_1, w_2\}$: each vector in the 1-simplex represents a credence function which assigns 0 to w_3 , and therefore, which is compatible with E ; so this bolded line is the ‘special subspace’ mentioned earlier. Finally, the dashed arrow from $\vec{c}$ to $\vec{c'}$ represents the map taking each vector in the 2-simplex to the vector in the 1-simplex with which the former vector is closest; think of this as a projection. So altogether, Close Update says the following: after agent A —with initial credences represented by vector $\vec{c}$ in the 2-simplex—obtains total centered world evidence represented by the 1-simplex E , A ’s new credences $\vec{c'}$ should be the projection of $\vec{c}$ into that 1-simplex.

It is an extremely attractive feature, of Close Update, that it admits of this interpretation. For geometric formulations of principles are often intrinsic, and explanatory, in a way that numerical formulations are not (Field, 1980). Geometric characterizations of rationality principles in particular, such as those based on the topologies of qualitative comparative confidence orderings, avoid problems faced by quantitatively numerical probability functions (Hawthorne, 2009; Savage, 1972), like problems arising from the infinite precisions of credences (Easwaran, 2015). Likewise, Close Update can be formulated using only the topological, geometric notions of vectors, simplexes, and projections – no appeals to numerical probabilities are needed; so the problems associated with those numerical formulations of rationality principles do not arise. Moreover, projections into subspaces, representing new total centered world evidence which agents have obtained, are natural ways to define update procedures. So this intrinsic and geometric formulation, with which Close Update is equivalent, amounts to a nice argument for Close Update.

2.2 Imaging, Indifference, and Minimal Revision

In this section, I explain how Close Update is equivalent to principles that, instead of accuracy, invoke imaging and indifference instead. These equivalences help elucidate the content, and attractiveness, of Close Update: for as these equivalences show, three plausible principles converge on the same account of updating. And after discussing this, I provide yet another argument for Close Update, based on the notion of a minimal revision.

Imaging is a procedure for transforming one credence function into another. The procedure involves taking the credences assigned to any given world, and shifting those credences to various other worlds. For a standard formalization of imaging (Fusco, 2023; Gärdenfors, 1982; Leitgeb, 2017; Lewis, 1976, 1981), let c be an agent's initial credence function, and let $E = \{w_1, \dots, w_k\}$ be that agent's total centered world evidence. Let T_E be the following function from pairs of worlds to $[0, 1]$.

$$T(w_i, w_j) = \begin{cases} 1 & \text{if } i \in \{1, \dots, k\} \text{ and } i = j \\ 0 & \text{if } i \in \{1, \dots, k\} \text{ and } i \neq j \\ \frac{1}{k} & \text{if } i \in \{k+1, \dots, n\} \text{ and } j \in \{1, \dots, k\} \\ 0 & \text{if } i, j \in \{k+1, \dots, n\} \end{cases}$$

Basically, $T_E(w', w)$ represents the percentage of credence shifted from w' to w . Then let c^{T_E} —call it the ' T_E -shift of c '—be the credence function such that for all worlds w , $c^{T_E}(w) = \sum_{w' \in \mathcal{W}} c(w')T_E(w', w)$. Each quantity $c(w')T_E(w', w)$ can be interpreted as the amount of $c(w')$ which gets shifted to w . And so $c^{T_E}(w)$ is the function resulting from all those shifts.

Imaging can be used to formulate an update principle for centered propositions. Basically, the principle says that after obtaining evidence E , an agent should update to a specific T_E -shift of their initial credence function.

Image Update

Let A be an agent. Let t be a time, and let t' be the time immediately after t . Let c and c' be the credence functions which A has at t and t' , respectively.

Suppose that at t' , A 's total evidence is some total centered world evidence E ; and let T_E be the shift function defined above. Then in order for A to be rational, c' should be c^{T_E} .

In other words, suppose that your total evidence is just your total centered world evidence. Then in order to rationally update your credences in response to that evidence, do the following. Consider the total amount of credence which your initial credence function assigns to worlds that your evidence rules out. Transfer those credences to worlds compatible with your evidence, in the way that the particularly nice shift function T_E specifies. According to Image Update, you should adopt the credence function which results.

As proved in the appendix, Close Update is equivalent to Image Update. And that counts in favor of both. For when theorizing about rationality, imaging principles like Image Update are extremely useful: they feature in causal decision theory (Joyce, 1999, p. 175; Lewis, 1981, pp. 14-15), defenses of group updating (Leitgeb, 2017, p. 463), theories of partial subjunctive supposition (Eva & Hartmann, 2021, p. 221; Zhang, 2024, p. 582), and forms of anti-Humeanism (Byrne & Hájek, 1997, p. 411). So imaging is a familiar, and fruitful, theoretical tool. And so the convergence between Close Update and Image Update tells in favor of both.⁹

Now for indifference. Basically, the following principle says that when updating, an agent should shift credences indifferently.

Indifferent Update

If an agent learns that some worlds do not obtain, and if there is nothing else which the agent learns, then the agent should adjust their credences in the following way: take the total credence which the agent had in the non-

⁹A technical aside: the proof of that convergence complements a well-known result due to Leitgeb and Pettigrew (2010b, pp. 243-244). As they showed, minimizing expected inaccuracy, using the squared Euclidean semimetric, is equivalent to an analog of Image Update. The main theorem proved in the appendix shows that simply minimizing closeness is equivalent to Image Update itself; nothing about expectation values—concerning accuracy or anything else—is needed. But interestingly, as shown by Snow Zhang, the function which Leitgeb and Pettigrew minimize is a linear shift of the function minimized in the main theorem; so in that sense, it is unsurprising that both approaches to minimization recommend adopting the same credence.

obtaining worlds, and divide it evenly among the worlds which remain.¹⁰

In other words, if an agent's total evidence is just their total centered world evidence, then the agent should (i) set their credences in the worlds eliminated by their evidence to zero, and (ii) increase their credence in each other world by the same fixed amount, where the amount in question depends on how much credence the agent had assigned to the worlds which were eliminated.

As established by theorems in the appendix, Close Update and Image Update are equivalent to Indifferent Update. And that supports two different arguments for Close Update. The first argument is based on the convergence of the three update principles. Each of Close Update, Image Update, and Indifferent Update, provide independently plausible accounts of how to update on total centered world evidence. The fact that these independently plausible accounts converge, in the sense of being mathematically equivalent—so that they deliver exactly the same verdicts in exactly the same cases—is evidence in favor of all three. In general, such convergences between independently plausible theories speaks in those theories' favor. Church makes this point when discussing how his preferred theory of effective computability, which employs the lambda calculus, is mathematically equivalent to Herbrand and Gödel's theory of effective computability based on recursiveness: the fact that “two such widely different and...equally natural definitions of effective calculability...turn out to be equivalent adds to the strength of the reasons” for endorsing them (1936, p. 346). Dyson makes a similar point, when establishing the equivalence of two different quantum theories of electrodynamics (1949, p. 486). Analogously, I claim, for Close Update and Image Update and Indifferent Update: the fact that three such different but equally natural theories—of updating on total centered world evidence—turn out to be equivalent, is a reason for endorsing them.

The second argument for Close Update is based on considerations about different ways to distribute credences among worlds. Suppose you are trying to decide how to divide up the credences, taken from the worlds which your new total centered world evidence rules out, among the worlds compatible with that

¹⁰Note that Indifferent Update, which places no constraints on agents' prior credences, is not susceptible to the problems faced by standard indifference principles (Rinard, 2014).

evidence. One candidate division is that which Indifferent Update, and therefore Close Update, recommends: divide evenly. All other candidate divisions are uneven – they privilege some worlds over others. Absent any evidential or mathematical reasons for privileging some worlds over others, however, such uneven distributions are unjustified. If worlds w_8 and w_9 are equally compatible with the evidence, for instance, and if the evidential context and background formal framework provide no reason whatsoever for privileging one of those worlds over the other, then all else equal, it would be irrational to give more credence to, say, w_8 than to w_9 .¹¹ Close Update, by endorsing an even division of credences, respects this. The denial of Close Update does not. And that is an argument in Close Update’s favor.

Another argument for Close Update is related to—but distinct from—the arguments just presented. It invokes the principle below.

Minimal Revision

If rationality requires that an agent revise their mental states, in order to satisfy a given collection of constraints, then the resulting mental states should—among all mental states satisfying those constraints—be as close as possible to the original mental states.

In other words, revisions in mental states should be minimal.

The minimization argument for Close Update is as follows. Suppose that an agent has a particular credence function, and suppose that the agent obtains new total centered world evidence. Then plausibly, rationality requires that the agent revise their mental states, in order to satisfy the following ‘elimination constraint’: the agent should adopt a credence function which assigns zero to all

¹¹Sometimes, all else is not equal – in which case, giving more credence to one world rather than another is perfectly rational. This happens in evidential contexts featuring uncentered propositions. In those contexts, agents need to update on evidence which is not centered, and so which does not repeatedly change truth value in the way that centered evidence does. As a result, plausibly, it is rational to give more credence to one world and less to another: worlds should be given credences in accord with preserving the prior credence ratios between them; and this requirement is well-defined only because there is no risk of a proposition, which the agent learned, later becoming false, in which case some credence ratios could correspond to division by zero. I discuss this more below, as well as in Section 2.3, where I continue exploring the relationship between conditionalization and Close Update in detail.

worlds that their new evidence rules out. Suppose, furthermore, that the agent's new total centered world evidence is also their new total evidence, full-stop. Then plausibly, there are no other constraints—beyond the elimination constraint—which the agent's new credences should satisfy. So by Minimal Revision, the agent's new credence function should be—among all credence functions satisfying the elimination constraint—as close as possible to the agent's original credence function. Along with the fact that closeness among credence functions is quantified using the squared Euclidean semimetric, Close Update follows.

Of course, to reject the minimization argument, one might reject Minimal Revision. But for several reasons, that would be costly; I discuss four here. First, if Minimal Revision is false, then it is rationally permissible for agents to shift their credences further than needed, in order to accommodate the relevant constraints. But there is no good reason to think that unnecessarily large shifts are rational. Unnecessarily large shifts are just that: unnecessary. So why do them? It is not clear what, exactly, justifies unnecessarily large shifts.

Second, justifying the rejection of Minimal Revision is, for subtle reasons, quite hard to do: basically any purported justification is, ultimately, a misapplication of what Minimal Revision says. To explain, suppose that there is indeed good reason for an agent to shift their mental states further than needed, in order to satisfy a given collection of constraints. Then plausibly, that good reason—for the unnecessarily large shift—is itself just another constraint, one which should have been included in the given collection of constraints to begin with. What appears to be an unnecessarily large shift, in other words, does not indicate a problem with Minimal Change. Rather, what appears to be an unnecessarily large shift indicates a problem with how Minimal Change was applied. The 'given collection of constraints', in Minimal Change, includes all the rationality constraints that apply to the agent in the case at issue. So if rationality requires that the agent shift to further-than-needed mental states, in order to accommodate a given collection of constraints, then plausibly, that collection omitted some rationality constraints which are relevant to the agent in question: the reasons for the further-than-needed shift, in other words, amount to constraints beyond those which the given collection included. And so the given collection of constraints was not, in fact, the collection of constraints to which Minimal Change should have been applied

in the first place.

Third, versions of Minimal Revision are endorsed throughout the literature. Several other ways of measuring closeness, designed for other sorts of evidential contexts than those in which the total evidence is centered, correspond to other updating principles. Conditionalization, for instance, does not distort probability ratios among certain uncentered propositions; so update by conditionalization is, in that sense, a form of minimal revision. Similarly, Jeffrey conditionalization minimizes the Kullback-Leibler divergence (Diaconis & Zabell, 1982; Meehan & Zhang, 2020). Close Update is similar, insofar as it preserves differences among credences in propositions which imply the given total centered world evidence: this follows from the equivalence between Close Update and Indifferent Update; and this supports Lewis’s point that imaging involves no gratuitous shifts of probability (1976, p. 311). The main upshot of all this: the minimization argument is simply based on another version of Minimal Revision. According to this other version, closeness among credences over centered propositions should be measured using the squared Euclidean semimetric. This version of Minimal Revision implies Close Update, just as other versions of Minimal Revision—for other evidential contexts, in which other minimal revisions are most appropriate—imply conditionalization and Jeffrey conditionalization.

Fourth, violations of Minimal Revision—and correspondingly, of Close Update—risk arbitrariness. According to a standard metatheoretical desiderata, philosophical principles should be non-arbitrary (Builes, 2022; Fairchild, 2022; Sider, 2007). If one principle delivers an arbitrary verdict, about a particular case, and another principle does not, then the latter is generally preferable to the former. Denials of Close Update, which endorse the permissibility of shifting credences further than the total centered world evidence requires, allow rational agents to adopt unattractively arbitrary credences. Minimal Revision and Close Update do not. And that is a strong point in favor of both.

To summarize: the equivalences discussed in this section support Close Update. Specifically, Close Update – formulated in terms of accuracy-based notions – is equivalent to Image Update – formulated in terms of imaging – which is equivalent to Indifferent Update – formulated in terms of indifference. These equivalences represent an elegant convergence in the theory of rationality being

pursued here. In addition, the notion of a minimal revision supports a compelling argument for Close Update as well. Rational update on centered evidence lies at the confluence of accuracy, imaging, indifference, and minimization.

2.3 Objections

In this section, I defend Close Update against four objections. The first and second concern conditionalization. The third concerns issues raised by Levinstein (2012). The fourth concerns Dutch Book arguments.

Regarding the first objection: one might worry that in some cases, conditionalization and Close Update conflict. For instance, suppose that for Sal, there are just three epistemically possible worlds: w_1, w_2 , and w_3 . In addition, suppose that initially, Sal's credences are such that $c(w_1) = \frac{1}{2}$, $c(w_2) = \frac{1}{4}$, and $c(w_3) = \frac{1}{4}$. Finally, suppose Sal learns that w_3 does not obtain. Then conditionalization would suggest that Sal's new credences should be $c'(w_1) = \frac{2}{3}$, $c'(w_2) = \frac{1}{3}$, and $c'(w_3) = 0$. Close Update, however, would suggest that Sal's new credences should be $c'(w_1) = \frac{5}{8}$, $c'(w_2) = \frac{3}{8}$, and $c'(w_3) = 0$. And this might seem like a problem for Close Update.

Not so. If the worlds in question are uncentered, then Close Update does not apply. And if the worlds in question are centered, then for many reasons, Sal's new credences should not be those given by conditionalization. Let me explain.

To start, suppose that w_1, w_2 , and w_3 are uncentered worlds. In other words, suppose that in the case at hand, the algebra of propositions—over which Sal's credences are defined—contains uncentered propositions only. Then the evidence which Sal receives, that rules out w_3 , is not total centered world evidence: for the evidence is not centered. So Close Update has no verdicts about what Sal's credences should be. And so there is no conflict between conditionalization and Close Update.

Alternatively, suppose that w_1, w_2 , and w_3 are centered worlds.¹² Then conditionalization would be a poor guide to how Sal's credences should change.

¹²As can be shown, either all of the worlds in a given relevant atomic algebra are centered or none of the worlds in that algebra are centered. This follows from various precise definitions of worlds, like the definition in Section 2.1.

To see why, suppose that the worlds are as follows.

1. w_1 : roses are red and it is Friday.
2. w_2 : roses are blue and it is Friday.
3. w_3 : roses are red and it is Saturday.

In the case at hand, Sal learns that w_3 does not obtain: perhaps Sal learns that it is Friday. So according to both conditionalization and Close Update, Sal's new credence function c' should be such that $c'(w_3) = 0$. But conditionalization—though not Close Update—implies that tomorrow, when Sal looks at a calendar and learns that it is Saturday, Sal's credence in w_3 is not rationally permitted to increase. And that is a problem: that credal increase is permissible.

Defenders of conditionalization sometimes respond, to examples like this, by insisting that the relevant worlds be indexed to times. For two different reasons, however, that insistence does not help here. First, indexing the above, centered worlds to times amounts to substituting uncentered worlds for centered worlds. For instance, indexing the world in which roses are red and it is Saturday to some fixed time t amounts to substituting an uncentered world, call it ' w_3^u ', for the centered world w_3 . And that amounts to changing the example, rather than responding to the example at issue. It is true that conditionalization works, as an update principle, for agents who learn uncentered propositions containing uncentered worlds like w_3^u . But I am asking a different question: what about agents who learn centered propositions containing centered worlds like w_3 ? Centered propositions and centered worlds—quite apart from their indexed-to-times, uncentered counterparts—play important roles in our doxastic lives. For all the reasons given by Lewis (1979), Perry (1979), and others, that role cannot be captured by theories which, like the present response on behalf of conditionalization, treat uncentered propositions only. So do not make the mistake of thinking that the issue, here, can be addressed by changing the worlds through some sort of indexing – that merely ignores the issue, sweeping it under the rug. The original issue remains: may agents, who learn centered propositions containing centered worlds like w_3 , rationally shift their credences from zero to non-zero values? Intuitively, the answer is: yes. Conditionalization, if used here, implies no; and that is a problem.

Second, there is a larger issue which this insistence, on indexing the relevant

worlds to times, does not resolve. Indexing worlds to times provides defenders of conditionalization with justification for claiming that the rational credences, in the relevant propositions, are never zero. But the larger issue is that intuitively, as cases like that involving Sal show, it is rationally permissible for credences in centered propositions to both be zero and also become non-zero later. Rationality allows for credences that both (i) are zero at one time, and (ii) are non-zero at another time. Indexing worlds to times makes it the case that, for the worlds in question, (i) does not occur. But in the intuitive case, which an account of updating should respect, (i) and (ii) *do* occur when the propositions in question are centered. In other words, as many intuitive examples involving centered propositions show, it is rationally permissible for agents' credences in a proposition to both (i) be zero, and then (ii) shift to a non-zero value; conditionalization does not accommodate that; and that is a reason to prefer Close Update for those centered propositions.

An important aside: this is related to why it would be a mistake to claim that uncentered propositions are numerically identical to certain disjunctions of centered worlds. It may well be true that some disjunctions of centered worlds are functionally equivalent, for certain confined purposes, to uncentered propositions: $w_1 \vee w_3$, for instance, may be—in a certain specific application of the above model to a certain agent at a certain time—treated as a reasonable stand-in for the proposition that roses are red. But literally identifying those disjunctions of centered worlds with uncentered propositions would be erroneous, for the reasons referenced above: the proposition $w_1 \vee w_3$ has a truth value which is constantly in flux, depending on what time it is, in a way which the proposition that roses are red is not. So the proposition $w_1 \vee w_3$ is centered, not uncentered.

The proper conclusion to draw, from all this, is that Close Update applies in one range of cases and conditionalization applies in another range of cases. The proper conclusion is not that conditionalization, or Close Update, should be rejected entirely. Rather, Close Update and conditionalization should both be endorsed. They just apply in different evidential contexts: the former in contexts involving centered propositions, and the latter in contexts involving uncentered propositions.¹³

¹³One might object by claiming that centered propositions are ‘merely representational’, and

Regarding the second objection: one might dislike the claim, just made, that there are two different update principles for two different kinds of cases. Intuitively, one might argue, there is something unattractively disunified about that sort of view. Better to have a single update principle for both uncentered propositions and centered propositions alike.

For many reasons, this objection fails; here, I discuss three. First, the objection is based on a bad intuition. There is an important, qualitative difference between cases involving uncentered propositions and cases involving centered propositions. There is a joint in nature between the rational reasoning patterns in which uncentered propositions figure and the rational reasoning patterns in which centered propositions figure; and we should accept that there is indeed such a joint, because of all the ways in which updating by conditionalization, on centered propositions, produces the intuitively incorrect results. Endorsing two update principles—one for the former patterns and one for the latter patterns—respects this joint, this qualitative difference. The objection does not.

Second, this objection is problematically unfriendly to the use of imaging in epistemology and decision theory. As explained in Section 2.2, Close Update is equivalent to Image Update. It is quite standard, in the literature, to use conditionalization to formulate principles about one range of cases, and to use imaging to formulate principles about another range of cases (Gärdenfors, 1982, p. 759; Leitgeb, 2017, p. 474). And that is all I am suggesting here: use conditionalization when theorizing about cases involving only uncentered propositions, and use Close Update—or equivalently, Image Update—when theorizing about cases involving centered propositions.

Third, this objection fails to respect the fact that sufficiently different evidential contexts require correspondingly different responses to evidence. An evidential context featuring only uncentered propositions is totally different, in all sorts of important ways, from an evidential context featuring centered propositions. And for the purposes of rational updating, evidential context matters. It matters whether the algebra, over which the agent’s credences are defined, features centered propositions or uncentered propositions, and it matters whether the evi-

so update principles for centered propositions should be constrained by update principles for uncentered propositions. In Section 3.2, I argue that this claim is incorrect.

dence received is total centered world evidence or something else entirely, because of how all that bears on the rationality—and even the mathematical coherence—of a given agent’s credences bouncing between zero and non-zero values. Clearly, such bouncing is rational in various evidential contexts; conditionalization is not even mathematically coherent in evidential contexts like those; so conditionalization simply cannot be the right update rule. To insist otherwise, and engage in the epicyclic process of proposing ever-more-convoluted twists on conditionalization to deal with the problems generated by evidential contexts featuring centered propositions, is to endorse a kind of ostrich Bayesianism. Better to breath air, not sand – centered propositions require a different update procedure, and there is nothing wrong with that.

Regarding the third objection: it might appear that Close Update has strange implications in cases like those which Levinstein discusses (2012, pp. 419-420). To illustrate, let c be Celie’s initial credence function, and consider the following centered worlds.

1. d_1 : dugongs eat seagrass and it is Monday.
2. $(\neg d)_1$: dugongs do not eat seagrass and it is Monday.
3. d_2 : dugongs eat seagrass and it is Tuesday.
4. $(\neg d)_2$: dugongs do not eat seagrass and it is Tuesday.

Suppose that initially, $c(d_1) = \frac{1}{100}$, $c((\neg d)_1) = \frac{99}{100}$, and $c(d_2) = c((\neg d)_2) = 0$. Then suppose that Celie goes to sleep, wakes up, looks at a calendar, and sees that it is Tuesday. So long as Celie’s total evidence is now the total centered world evidence $\{d_2, (\neg d)_2\}$, Close Update implies that in order to be rational, Celie’s new credence function c' should be such that (i) $c'(d_1) = c'((\neg d)_1) = 0$, and (ii) $c'(d_2) = c'((\neg d)_2) = \frac{1}{2}$. And that, one might think, is unintuitive: Celie is not rationally required to shift her credences from (i) being almost completely confident in the proposition “Dugongs eat seagrass,” to (ii) being equally confident in “Dugongs eat seagrass” and “Dugongs do not eat seagrass.”

The source of the unintuitiveness, however, is not Close Update. The source is the relationship between (i) the total evidence which, intuitively, Celie has after learning that it is Tuesday, and (ii) the total centered world evidence in the model above, namely $\{d_2, (\neg d)_2\}$. For intuitively, after learning that it is Tuesday, Celie’s total evidence includes more than just d_2 and $(\neg d)_2$. Intuitively, Celie’s

total evidence also includes propositions about what her credences used to be, since she remembers that yesterday, she was almost completely sure that dugongs do not eat seagrass and it is Monday. This fact, about what Celie remembers, is not represented in the model above. So in the case involving Celie, the relevant total evidence includes more than Celie's total centered world evidence $\{d_2, (\neg d)_2\}$. Close Update only applies, however, when the relevant total evidence is the agent's total centered world evidence; so Close Update does not apply in this case. And moreover, when the additional evidence concerning Celie's memory is accommodated in the right way, the result is that when Celie looks at a calendar and sees that it is Tuesday, her new credences should be shifts of her old credences: the new credence function c'' which Celie should have, in other words, is such that $c''(d_2) = \frac{1}{100}$ and $c''((\neg d)_2) = \frac{99}{100}$.¹⁴

Regarding the fourth objection: one might worry that Close Update is incompatible with Dutch Book arguments. For some Dutch Book arguments show that given several substantive assumptions about rational credences and the truth values of propositions, agents who do not conform to conditionalization will always lose money in certain situations. Since rational update principles which generate forced monetary losses are unattractive, one might claim, Close Update is unattractive too.

For straightforward reasons, this objection fails. Several substantive assumptions, invoked in Dutch Book arguments for conditionalization, are false in cases involving centered propositions (Lewis, 1999, pp. 404-405). For instance, one as-

¹⁴Though an in-depth discussion is beyond the scope of this paper, it is still worth sketching one of the several ways in which Close Update can be used to derive this. As mentioned earlier, c' is the closest credence function to c compatible with $\{d_2, (\neg d)_2\}$. Now, on Tuesday, Celie remembers the following: (i) her credence in d_1 was $\frac{1}{100}$ the day before, and (ii) her credence in $(\neg d)_1$ was $\frac{99}{100}$ the day before. In addition, Celie knows that her total centered world evidence does nothing other than rule out certain times: that is, Celie knows that her total evidence $\{d_2, (\neg d)_2\}$ says merely that it is not Monday. Therefore, according to the current proposal, there is a special association which Celie should draw between d_1 and d_2 , as well as between $(\neg d)_1$ and $(\neg d)_2$. Intuitively, the purpose of this association is to capture Celie's memory of her past credences, as well as Celie's knowledge that her total evidence merely shifts the time by a day. Formally, the purpose of this association is to provide a guide as to which credences are weighed by which other credences. For according to the current proposal, the credence function c'' , which Celie should adopt after learning $\{d_2, (\neg d)_2\}$, is defined as follows. First, $c''(d_1) = c''((\neg d)_1) = 0$, since those centered worlds are ruled out by Celie's total evidence. Second, $c''(d_2) = c'(d_2)c(d_1)/n = \frac{1}{200} \cdot \frac{1}{n} = \frac{1}{100}$ and $c''((\neg d)_2) = c'((\neg d)_2)c((\neg d)_1)/n = \frac{99}{200} \cdot \frac{1}{n} = \frac{99}{100}$, where $n = c'(d_2)c(d_1) + c'((\neg d)_2)c((\neg d)_1) = \frac{1}{2}$ is the normalization factor.

sumption is that rational agents' initial credence functions must assign non-zero credences to every proposition which could become those agents' total evidence. This does not hold in cases involving centered propositions because often, rational agents assign zero credence to propositions which—like “It is Saturday”—could become those agents' total evidence at some point in the future.¹⁵

2.4 Chance

There is another strong argument for Close Update: when extended in the natural way to accommodate chances, Close Update implies popular verdicts for both a case discussed by Elga (2000)¹⁶ and the Judy Benjamin case (van Fraassen, 1981). In this section, I describe that extension of Close Update. Then I explain how that extension has those implications.

To start, some preliminaries. Let ch be a function – call it a ‘chance distribution’ – which maps each world w in $\mathcal{W}$ to a chance function ch_w – call it the ‘chance function that obtains at w ’ – which assigns chances to the worlds that $\mathcal{W}$ contains.¹⁷ Then say that an agent's ‘total centered chance evidence’ is a pair $\langle E, ch \rangle$ where (i) E is the agent's total centered world evidence, and (ii) ch is a chance distribution. For simplicity, assume that for each $\langle E, ch \rangle$, each $w \in E$, and each ch_w , $ch_w(e) > 0$; where e is the disjunction of all worlds that E contains.

Here is the extension of Close Update. Basically, it says that after obtaining new total centered chance evidence, an agent should update to a credence function that is a weighted average of the chance functions which obtain at the worlds with which that evidence is compatible.

Chancy Close Update

Let A be an agent. Let c be the credence function which A has at time t , and let c'' be the credence function which A has at the time t'' immediately after

¹⁵In fact, there are even Dutch Book arguments in favor of imaging (Zhang, in preparation), which in turn support Close Update.

¹⁶For another approach to Elga's case which employs accuracy, compatible with though different from mine, see (Kierland & Monton, 2005).

¹⁷For an introduction to this formalism, see (Dorst, 2019; Gallow, 2023; Williamson, 2008).

t . Suppose that at t'' , A comes to have new total centered chance evidence $\langle E, ch \rangle$. Let c' be the closest credence function to c which conforms to E . Let $ch_1, \dots, ch_m$ be the list of all the chance functions to which ch maps some world or other in E , such that no chance function occurs twice in the list. For each i such that $1 \leq i \leq m$, let E_i be the set of all worlds in E at which ch_i is the chance function, and let e_i be the disjunction of all worlds which E_i contains.¹⁸ Then in order for A to be rational, c'' should be the function defined below.

$$c''(w) = \begin{cases} \sum_{i=1}^m \frac{c'(e_i)}{ch_i(e)} ch_i(w) & \text{if } w \text{ is in } E \\ 0 & \text{otherwise} \end{cases}$$

In other words, after learning $\langle E, ch \rangle$, an agent's new credence function should have two features.¹⁹ First, the new credence function should assign zero to all worlds which E rules out. Second, when it comes to worlds in E , the new credence function should be the weighted average of the chance functions which obtain at those worlds, where the weights are given by whatever credence function which conforms to E is closest to the agent's original credence function.²⁰

Now for Elga's case. On Sunday, Susie is told the following. She will be put to sleep that evening. Then she will be woken on Monday. After a brief period of time, she will learn that it is Monday. Then she will be put back to sleep. Later that evening, a fair coin will be flipped. If the coin lands heads, she will be woken on Wednesday and the experiment will be over. If the coin lands tails, she will be given a special drug which erases her memory of the Monday awakening, and she will be woken on Tuesday. Then she will be put back to sleep, she will be woken on Wednesday, and the experiment will be over. This is Susie's experiment.

In Susie's experiment, there are six possible ways that the world could be. They are as follows.

¹⁸More rigorously, consider the equivalence relation $\sim$ defined like this: for all $v, w \in E$, $v \sim w$ if and only if $ch_v = ch_w$. This induces a partition $P_E = \{E_i\}_{i=1}^m$ on E (for some $m \in \{1, \dots, k\}$ where $k = |E| \leq n = |\mathcal{W}|$). Finally, for each $E_i \in P_E$, let ch_i be the chance function which the worlds in E_i share in common.

¹⁹Since $\sum_{w \in \mathcal{W}} c''(w) = \sum_{w \in E} \sum_{i=1}^m \frac{c'(e_i)}{ch_i(e)} ch_i(w) = \sum_{i=1}^m c'(e_i) \sum_{w \in E} \frac{ch_i(w)}{ch_i(e)} = \sum_{i=1}^m c'(e_i) = 1$, and since $c''(w) = \sum_{i=1}^m \frac{c'(e_i)}{ch_i(e)} ch_i(w) \geq 0$ for each $w \in E$, c'' is a probability function over $\mathcal{W}$.

²⁰For principles somewhat similar to Chancy Close Update, see (Hall, 1994; Ismael, 2008).

h_{su} : the coin lands heads and it is Sunday.
 t_{su} : the coin lands tails and it is Sunday.
 h_{mo} : the coin lands heads and it is Monday.
 t_{mo} : the coin lands tails and it is Monday.
 h_{tu} : the coin lands heads and it is Tuesday.
 t_{tu} : the coin lands tails and it is Tuesday.

On Sunday, Susie's credence function c is as follows.

$$c(w) = \begin{cases} \frac{1}{2} & \text{if } w \text{ is in } \{h_{su}, t_{su}\} \\ 0 & \text{otherwise} \end{cases}$$

Now ask: after Susie wakes on Monday, but before she learns that it is Monday, how confident should she be in the proposition, namely $h_{su} \vee h_{mo} \vee h_{tu}$, that the coin lands heads? Many—and perhaps even most—argue for the ‘thirder view’: Susie should have credence $\frac{1}{3}$ in that proposition at that time (Elga, 2000; Weatherson, 2013; Weintraub, 2004; Wilhelm, 2021).²¹

Chancy Close Update can be used to formulate an argument for the thirder view. In order to do so, each possibility h_{su} , t_{su} , h_{mo} , t_{mo} , h_{tu} , and t_{tu} , must be associated with a chance function. As it turns out, for many natural choices of chance functions to associate with worlds in $\mathcal{W}$, Chancy Close Update implies that Susie is rationally required to have the same credences. Just to keep things simple, assume that for every w in $\mathcal{W}$, the chance function ch_w assigns $\frac{1}{6}$ to each of h_{su} , t_{su} , h_{mo} , t_{mo} , h_{tu} , and t_{tu} ; so every world is associated with the very same chance function.

From this, the thirder view follows. For after waking on Monday but before being told that it is Monday, Susie has centered evidence $E = \{h_{mo}, t_{mo}, t_{tu}\}$. As shown by a straightforward application of theorem 1 in the appendix, the closest credence function to c which conforms to E is the function defined below.

$$c'(w) = \begin{cases} \frac{1}{3} & \text{if } w \text{ is in } \{h_{mo}, t_{mo}, t_{tu}\} \\ 0 & \text{otherwise} \end{cases}$$

²¹For arguments in favor of alternative views, see (Lewis, 2001; White, 2006).

This, along with Chancy Close Update, implies that in order to be rational, Susie's credence function should be as follows.

$$c''(w) = \begin{cases} \frac{1}{3} & \text{if } w \text{ is in } \{h_{mo}, t_{mo}, t_{tu}\} \\ 0 & \text{otherwise} \end{cases}$$

So when Susie wakes up on Monday, but before she is told that it is Monday, she should have credence $\frac{1}{3}$ in the proposition $h_{su} \vee h_{mo} \vee h_{tu}$ that the coin lands heads. Hence the third view.

This is a reason to endorse Chancy Close Update, and correspondingly, Close Update. For when extended in a natural way, to accommodate cases involving chance, Close Update implies perhaps the most popular view of rational credences in Elga's case. So Close Update is worth endorsing.

Now for the Judy Benjamin case. Private Judy, who has just parachuted into unknown land, has an initial credence function c which assigns $\frac{1}{4}$ to each of the following four worlds.

- $r \wedge s$: I am in the red territory and the Second Company area.
- $r \wedge \neg s$: I am in the red territory and the Headquarters Company area.
- $\neg r \wedge s$: I am in the blue territory and the Second Company area.
- $\neg r \wedge \neg s$: I am in the blue territory and the Headquarters Company area.

Then Judy learns the following over radio: the conditional chance that I am in the Headquarters Company area, given that I am in the red area, is $\frac{3}{4}$. As discussed by Douven and Romeijn, the challenge in the Judy Benjamin case is to propose an update rule which—on the basis of what Judy learned—implies that Judy's final credence function c'' satisfies the following constraints (2011, p. 639):

- (i) $c''(\neg s \mid r) = \frac{3}{4}$,
- (ii) for all propositions y , and for all x in $\{r \wedge s, r \wedge \neg s, \neg r\}$, $c''(y \mid x) = c(y \mid x)$,
and
- (iii) $c''(\neg r) = c(\neg r) = \frac{1}{2}$.

Strikingly, given the most plausible choice of a chance distribution ch which Judy learns about from the radio, Chancy Close Update implies that c'' obeys the four constraints above.²² This is an extremely attractive feature of Chancy Close Update.

An attractive upshot: all this helps accuracy-based approaches to rationality avoid a dilemma discussed in the literature (Briggs, 2010; Builes, 2020). The dilemma is that because different expected accuracy minimization procedures yield different rationality recommendations in Elga’s case—some procedures imply the third view; others do not—that case cannot be settled by accuracy-based approaches which invoke expected accuracy claims. As discussed in Section 2.2, however, Close Update—despite being an accuracy-based approach—does not make any claims about the expected accuracy of anything. Close Update only makes claims about how close one credence function is to another credence function. That is why, as was just shown, extensions of Close Update can be used to argue for the third view. So Close Update is an accuracy-based approach to rationality which avoids a dilemma that other accuracy-based approaches face.

3 Comparisons

In this section, I compare Close Update with alternative theories of updating in the literature. To start, I argue that Close Update enjoys some advantages over three other principles for updating on centered propositions. Then I discuss a pervasive but overlooked assumption, in the literature on updating, which Close Update challenges.

²²In rough outline, here is how. Given the content of the radio message, the best choice for ch maps each of the four worlds listed earlier to the same chance function ch_1 defined as follows: $ch_1(\neg s \mid r) = \frac{3}{4}$, as this is what the radio message conveyed, and $ch_1(\neg r \wedge s) = ch_1(\neg r \wedge \neg s) = \frac{1}{4}$, as no information was provided about the chances of Judy being in the blue territory. From all this, it follows that $ch_1(r \wedge s) = \frac{1}{8}$ and $ch_1(r \wedge \neg s) = \frac{3}{8}$. And an elementary but tedious proof shows that this, together with the fact that the radio message did not rule out any of the four worlds mentioned earlier—and so the best choice for E includes all of those worlds—implies, by Chancy Close Update, that c'' satisfies those four constraints

3.1 Advantages Over Alternatives

The update principles, with which I compare Close Update, are due to Meacham, Moss, and Titelbaum. My remarks apply to other principles too, such as those in (Isaacs et al., 2022). But for lack of space, I do not discuss that here.

A variant of the case involving Jie can be used to illustrate an advantage, which Close Update has, over Meacham’s ‘Compartmentalized Conditionalization’ principle (2008). For each month and day in that month, consider all centered worlds of the form either “Dugongs eat seagrass and it is that month and day of 2025” or “Dugongs do not eat seagrass and it is that month and day of 2025.” Suppose that initially, Bell has credence $\frac{1}{730}$ in each of these worlds; so Bell has credence $\frac{1}{2}$ in dugongs eating seagrass, and Bell has credence $\frac{1}{2}$ in dugongs not eating seagrass. Then suppose that Bell’s total evidence becomes the total centered world evidence E containing (i) the world “Dugongs eat seagrass and it is January 1 of 2025,” (ii) every world of the form “Dugongs do not eat seagrass and it is that month and that day of 2025,” and (iii) nothing else.

Compartmentalized Conditionalization delivers an implausible verdict in this case. According to Compartmentalized Conditionalization, Bell’s new credences in two propositions—that dugongs eat seagrass, and that dugongs do not eat seagrass—cannot change when E becomes her total evidence; the reason concerns the fact that those propositions, in the algebra at issue, are functionally equivalent to uncentered propositions. So as some calculations show, Compartmentalized Conditionalization implies that Bell’s new credence in “Dugongs eat seagrass and it is January 1 of 2025” should be $\frac{1}{2}$, and Bell’s new credence in each world of the form “Dugongs do not eat seagrass and it is thus-and-so month and day of 2025” should remain at $\frac{1}{730}$. But that is implausible. For it amounts to claiming that Bell’s credence in “Dugongs eat seagrass and it is January 1 of 2025” should increase by a factor of 365, while Bell’s credences in all other centered worlds should not increase at all.

Close Update delivers a much more plausible verdict in this case. For Close Update allows Bell’s new credences in the two propositions mentioned earlier—that dugongs eat seagrass, and that dugongs do not eat seagrass—to change when E becomes her total evidence. As given by theorem 1 in the appendix, Close Up-

date implies that Bell's new credence in each epistemically possible world should be $\frac{1}{366}$. And that is perfectly reasonable.

Close Update also enjoys several advantages over update principles due to Moss (2012) and Titelbaum (2008). For starters, those other update principles do not cover cases which, like Elga's case and Judy Benjamin's case, involve chances. They do not say how chances should affect such updating. Chancy Close Update, a natural extension of Close Update, does; and that is a significant reason to prefer Close Update.

In addition, Close Update avoids certain ambiguity problems that arise for those other update principles. For instance, take the principle 'Proxy' endorsed by Moss (2012). According to Proxy, for every agent with some total evidence, and for every centered proposition, there is an uncentered proposition which—for the purposes of updating—can serve as the centered proposition's 'proxy': for the purposes at hand, that uncentered proposition is equivalent to that centered proposition. The ambiguity problem is that in many cases, it is not clear what the proxy of the given centered proposition is. For instance, no uncentered proposition seems suitable to serve as a proxy for "Dugongs eat seagrass and it is January 1 of 2025" in the case involving Bell.

Similarly, take the 'Proper Expansion Principle' which Titelbaum endorses (2008). Very roughly put, according to this principle, rational agents should update in a way that respects certain expansions of the algebra of propositions over which those agents' credences are defined. The ambiguity problem is that in many cases, there are multiple inequivalent expansions of the algebra of propositions at issue. These expansions generate different verdicts about how the relevant agent's credences should evolve. And it is not always clear which verdicts are the right ones: the rationality verdicts, which the Proper Expansion Principle yields, are pretty unconstrained.

Close Update avoids both problems. For Close Update does not rely on anything like an ambiguous, undefined relation ' $\dots$ is a proxy for $\dots$ '; nor does Close Update rely on anything like ambiguous, underspecified guidelines for expanding algebras. To use Proxy, we must make choices about proxies which Moss's theory leaves largely unconstrained. To use the Proper Expansion Principle, we must make choices about algebra expansions which Titelbaum's theory leaves largely

unconstrained. But to use Close Update, we need not make any such unconstrained choices like those. Put another way: while Close Update, Moss’s theory, and Titelbaum’s theory, model rational updates by invoking evidence and initial credences, only Close Update avoids invoking other free parameters which resist precise specification. And that is a point in favor of Close Update, because in any given case, there is room to disagree over what the imprecise free parameters’ values—the proxies, the algebra expansions, or whatever—are.

3.2 The Orthodox Assumption

It is worth stepping back from comparisons with specific principles in the literature, and instead discussing a more general assumption—call it the ‘orthodox assumption’—which underwrites those principles, and which Close Update contradicts. According to the orthodox assumption, updating on uncentered propositions is, in some important sense or other, more basic than updating on centered propositions. Perhaps update principles for centered propositions should be constrained by, or reduced to, or defined in terms of, principles concerning uncentered propositions only. Regardless, the orthodox assumption claims that uncentered propositions play a privileged role in determining which updates on centered propositions are rational, and which are not.²³

Close Update contradicts all that. For Close Update does not require that updating on centered propositions be constrained by, or reduced to, updating on uncentered propositions. According to Close Update, rational updating on centered propositions is not some sort of secondary, anomalous case of rational updating on uncentered propositions. Given Close Update, uncentered propositions do not play a privileged role in determining which updates on centered propo-

²³This assumption is so common that there are few, if any, explicit statements of it. But Meacham’s claim that ratios among credences in certain uncentered propositions should be preserved upon learning centered propositions (2008, p. 249), and Moss’s claim that every centered proposition has an uncentered proxy (2012, p. 226), are examples of the orthodox assumption at work. And more generally, philosophers often stipulate that principles of rationality should be impersonal, and independent of self-location, in ways that amount to endorsing the orthodox assumption too: see Hedden’s claim that “the requirements of rationality should be impersonal” (2015, p. 452); see also (Builes, 2020, p. 3034; Christensen, 2000, p. 363).

tions are rational, and which are not. Rational updating on centered propositions concerns total evidence, and closeness among credences, only.

One might try to argue for the orthodox assumption—and therefore against Close Update—by citing a popular view of what uncentered propositions, and centered propositions, represent. According to this view, uncentered propositions represent genuine ways for the world to be, while centered propositions do not. Instead, centered propositions merely represent ways to be located within the world, where ‘ways to be located within the world’ are not genuine ways for the world to be. For endorsements of something like this, see Elga: “[i]n addition to being uncertain about what the world is like, one can also be uncertain about one’s own spatial or temporal location in the world” (2000, p. 143). Or see Bradley: “[s]ometimes we learn what the world is like. Other times, we learn about where we are in the world” (2011, p. 393). By ‘what the world is like’, these authors mean an uncentered proposition, and by ‘location’—or by ‘where we are in the world’—they mean a centered proposition. So these quotes suggest that uncertainty about a centered proposition, or learning a centered proposition, does not amount to uncertainty about—or learning—a genuine way for the world to be. And that supports the orthodox assumption. For it is natural to think that uncertainty about how the world is—uncertainty, in other words, about uncentered propositions—should play a special, privileged role in all other sorts of uncertainty, including uncertainty about centered propositions. Hence orthodoxy.

My response: centered propositions do represent genuine ways for the world to be. It is true, of course, that centered propositions represent ways to be located within the world. But ways to be located within the world are genuine ways for the world to be. Where I am, in the world, is a way that the world is. The time now is a way that the world is too. There is no hard distinction between (i) being uncertain about what the world is like, and (ii) being uncertain about where we are in the world. Neither represents ‘genuine’ uncertainty, about the world, any more than the other does. Both represent fully genuine kinds of uncertainty about what the world is like.

Here is another way to put the point. Contrary to orthodoxy, what you should believe does, indeed, depend on who you are. Of course it does. It is, after all, what *you* should believe. And of course that depends on *you*. What

you should believe differs from what *I* should believe; so what you should believe depends on who you are, just as what I should believe depends on who I am. Similarly, what you should believe *now* differs from what you should believe *then*; so what you should believe now depends on what time it is now, just as what you should believe then depends on the time then. So the orthodox assumption is worth rethinking – and so it is important to explore principles which, like Close Update, do not presuppose the orthodox assumption automatically.

4 Conclusion

This paper features many different arguments for Close Update. Here are some of them: Close Update

- (i) implies the intuitively correct verdicts in various cases,
- (ii) is equivalent to an updating principle based on imaging,
- (iii) is equivalent to an updating principle based on indifference,
- (iv) follows from a plausible minimization postulate,
- (v) is preferable to conditionalization when it comes to updating on centered propositions,
- (vi) extends to Chancy Close Update, which can be used to derive the third view in Elga’s case and the intuitive verdicts in the Judy Benjamin case,
- (vii) avoids a dilemma which accuracy-based approaches generally face,
- (viii) enjoys several advantages over alternative update principles, and
- (ix) challenges the literature’s problematic tendency to treat centered propositions as secondary to uncentered propositions.

So Close Update has a lot in its favor. It is worth taking seriously.

Appendix

In this appendix, I prove some results mentioned in the main text. Let $\mathcal{W}$ have fixed enumeration $\mathcal{W} = \{w_1, \dots, w_n\}$. Let $\mathcal{C}$ be the set of credence functions

over $\mathcal{W}$. For each $w \in \mathcal{W}$, let $1_w : \mathcal{W} \rightarrow [0, 1]$ be the following function in $\mathcal{C}$: $1_w(v) = 1$ if $v = w$ and $1_w(v) = 0$ otherwise. Let $E \subseteq \mathcal{W}$ have fixed enumeration $E = \{w_1, \dots, w_k\}$, and define

$$\text{conv}(E) = \left\{ c \in \mathcal{C} \mid \exists \lambda_1, \dots, \lambda_k \in [0, 1] \text{ s.t. } \sum_{i=1}^k \lambda_i = 1, c = \sum_{i=1}^k \lambda_i 1_{w_i} \right\} \quad (1)$$

For all $c, c' \in \mathcal{C}$, the degree of closeness between c and c' is given by the squared Euclidean semimetric:

$$\mathfrak{s}(c, c') = \sum_{i=1}^n (c(w_i) - c'(w_i))^2 \quad (2)$$

For all $c \in \mathcal{C}$ and $E \subseteq \mathcal{W}$, say that c ‘conforms to E ’ just in case $c \in \text{conv}(E)$. For all $c \in \mathcal{C}$ and all $E \subseteq \mathcal{W}$, $c' \in \mathcal{C}$ is the ‘closest credence function to c which conforms to E ’ just in case (i) $c' \in \text{conv}(E)$, and (ii) for all $c'' \in \text{conv}(E)$, $\mathfrak{s}(c, c') \leq \mathfrak{s}(c, c'')$.

The following lemma provides an explicit expression for the function c^{T_E} , which I called the ‘ T_E -shift of c ’ in Section 2.2.

Lemma 1. *For each $j \in \{1, \dots, k\}$, $c^{T_E}(w_j) = \frac{1}{k} \left(1 - \sum_{i=1}^k c(w_i) \right) + c(w_j)$. And for each $j \in \{k+1, \dots, n\}$, $c^{T_E}(w_j) = 0$.*

Proof. For each $j \in \{1, \dots, k\}$, $c^{T_E}(w_j) = \sum_{i=1}^n c(w_i) T_E(w_i, w_j) = \sum_{i=k+1}^n c(w_i) T_E(w_i, w_j) + \sum_{i=1}^k c(w_i) T_E(w_i, w_j) = \sum_{i=k+1}^n \frac{1}{k} c(w_i) + c(w_j) + \sum_{i=1, i \neq j}^k c(w_i) T_E(w_i, w_j) = \frac{1}{k} \left(1 - \sum_{i=1}^k c(w_i) \right) + c(w_j)$. Similarly, for each $j \in \{k+1, \dots, n\}$, $c^{T_E}(w_j) = \sum_{i=1}^n c(w_i) T_E(w_i, w_j) = \sum_{i=k+1}^n c(w_i) T_E(w_i, w_j) + \sum_{i=1}^k c(w_i) T_E(w_i, w_j) = 0 + 0 = 0$. $\square$

Here is the main theorem of this appendix. It says that Close Update is equivalent to an indifference requirement on updating.

Theorem 1. *Let $\mathcal{W}$, $\mathcal{C}$, E , and c be as defined above. Then c' is the closest credence function to c which conforms to E if and only if for all $j \in \{1, \dots, k\}$, $c'(w_j) = \frac{1}{k} \left(1 - \sum_{i=1}^k c(w_i) \right) + c(w_j)$.*

Proof. By (1),

$$\text{conv}(E) = \left\{ c^* \in \mathcal{C} \mid \exists \lambda_1, \dots, \lambda_k \in [0, 1] \text{ s.t. } \sum_{i=1}^k \lambda_i = 1, c^* = \sum_{i=1}^k \lambda_i 1_{w_i} \right\}$$

So for each $c^* \in \text{conv}(E)$, there are $\lambda_1, \dots, \lambda_k \in [0, 1]$ such that $\lambda_k = 1 - \sum_{i=1}^{k-1} \lambda_i$ and $c^* = \left(\sum_{i=1}^{k-1} \lambda_i 1_{w_i}\right) + \left(1 - \sum_{i=1}^{k-1} \lambda_i\right) 1_{w_k}$. In addition, note that there are $c_1, \dots, c_n \in [0, 1]$ such that $\sum_{i=1}^n c_i = 1$ and $c = \sum_{i=1}^n c_i 1_{w_i}$. All that, along with (2), implies that for each $c^* \in \text{conv}(E)$, $\mathfrak{s}(c, c^*) = \sum_{i=1}^n (c(w_i) - c^*(w_i))^2 = \left(\sum_{i=1}^{k-1} (c_i - \lambda_i)^2\right) + \left(c_k - \left(1 - \sum_{i=1}^{k-1} \lambda_i\right)\right)^2 + \sum_{i=k+1}^n c_i^2$. The closest credence function to c , among all credence functions which conform to E , is the function $c' \in \text{conv}(E)$ that, when substituted for c^* , minimizes $\mathfrak{s}(c, c^*)$. So among all credence functions which conform to E , the closest to c is the one whose corresponding $\lambda_1, \dots, \lambda_{k-1}$ minimize the function $f(\lambda_1, \dots, \lambda_{k-1}) = \left(\sum_{i=1}^{k-1} (c_i - \lambda_i)^2\right) + \left(c_k - \left(1 - \sum_{i=1}^{k-1} \lambda_i\right)\right)^2 + \sum_{i=k+1}^n c_i^2$. To establish both directions of the theorem, it suffices to find the values $\lambda_1, \dots, \lambda_{k-1}$ which minimize $f(\lambda_1, \dots, \lambda_{k-1})$. From calculus, $f(\lambda_1, \dots, \lambda_{k-1})$ is minimized by some values $\lambda'_1, \dots, \lambda'_{k-1}$ such that for all $j \in \{1, \dots, k-1\}$, when $\lambda'_1, \dots, \lambda'_{k-1}$ are plugged into $\frac{\partial f}{\partial \lambda_j}$, the result is 0. So to find those $\lambda'_1, \dots, \lambda'_{k-1}$, note the following: for all $j \in \{1, \dots, k-1\}$, $\frac{\partial f}{\partial \lambda_j} = -2(c_j - \lambda_j) + 2\left(c_k - \left(1 - \sum_{i=1}^{k-1} \lambda_i\right)\right)$. Therefore, for all $j \in \{1, \dots, k-1\}$, $\frac{\partial f}{\partial \lambda_j} = 0$ if and only if $\lambda_j - c_j + c_k - 1 + \sum_{i=1}^{k-1} \lambda_i = 0$, in other words, if and only if $\lambda_j = 1 - \left(\sum_{i=1}^{k-1} \lambda_i\right) - c_k + c_j = \lambda_k - c_k + c_j$. So the values $\lambda_1, \dots, \lambda_{k-1}$ which minimize $f(\lambda_1, \dots, \lambda_{k-1})$ have the following feature: for each $j \in \{1, \dots, k-1\}$,

$$\lambda_j - c_j = \lambda_k - c_k \quad (3)$$

Furthermore, note that $\lambda_k - c_k = \frac{1}{k}k(\lambda_k - c_k) = \frac{1}{k} \sum_{j=1}^k (\lambda_k - c_k) = \frac{1}{k} \sum_{j=1}^k (\lambda_j - c_j) = \frac{1}{k} \sum_{j=1}^k \lambda_j - \frac{1}{k} \sum_{j=1}^k c_j = \frac{1}{k} \left(1 - \sum_{i=1}^k c_i\right)$, where the third equality follows from (3). And this, along with (3), implies that for all $j \in \{1, \dots, k\}$, $\lambda_j = \frac{1}{k} \left(1 - \sum_{i=1}^k c_i\right) + c_j$. Therefore, the closest credence function in $\text{conv}(E)$ to c is the function $c' \in \text{conv}(E)$ such that for all $j \in \{1, \dots, k\}$, $c'(w_j) = \lambda_j = \frac{1}{k} \left(1 - \sum_{i=1}^k c_i\right) + c_j = \frac{1}{k} \left(1 - \sum_{i=1}^k c(w_i)\right) + c(w_j)$. $\square$

Corollary 1. *Let $\mathcal{W}$, $\mathcal{C}$, E , c , and T_E be as defined above. Then c' is the closest credence function to c which conforms to E if and only if $c' = c^{T_E}$.*

Proof. Follows from lemma 1 and theorem 1. $\square$

Theorem 1 and corollary 1 show that Close Update, Image Update, and Indifferent Update are equivalent.

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